

Proteomics Analysis Using the JMod GUI

While JMod can be run from the command line, many users may find it easier to use a graphical user interface. Here, we walk through the important aspects of the user interface and how it can be used to analyze data with JMod.

Installation:

JMod can be downloaded as an executable for [hyperlink\[windows\]](#) or [hyperlink\[mac\]](#), or found on [GitHub](#).

If using windows, it is recommended to enable long paths to ensure all outputs are properly saved. To do this:

- 1) Press Win+R
- 2) Type 'regedit' and hit enter
- 3) Navigate to HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\FileSystem
- 4) Set LongPathsEnabled to 1
- 5) Restart computer

Processing the data:

JMod requires mzML files as inputs. For conversion from Thermo .raw files, we recommend using msConvert with parameters:

- Peak Picking = 1-
- The rest are default

Working with the GUI:

Step 1: Input mzML file(s) and spectral library and select presets

The screenshot displays the JMod GUI interface. The 'Input Files' section is highlighted with a red box, showing a file selection button and a 'Browse' button. Below this, the 'Spectral Library' is set to 'C:/Users/zcohe/mod/Random Data/Splex/diann_tag6_Astral_MBI' with a 'Browse' button. The 'Presets' section shows 'PSMtag_Splex_DIA_Defaults.json' with a 'Browse' button. The 'MS Settings' section includes 'MS1 ppm' (0.0), 'Min Fragments' (3), 'RT Tolerance' (0.5), 'MS2 ppm' (10.0), 'Min Lib Match' (0.5), 'MS2 Isotopes' (3), and a 'Use Lib RT' checkbox. The 'Multiplexing' section shows a 'Tag' dropdown set to 'PSMtag_Splex' and a 'Timeplex' dropdown set to '0'. A 'Create New Tag' button is present. The 'Mass Tag Info' table shows the following data:

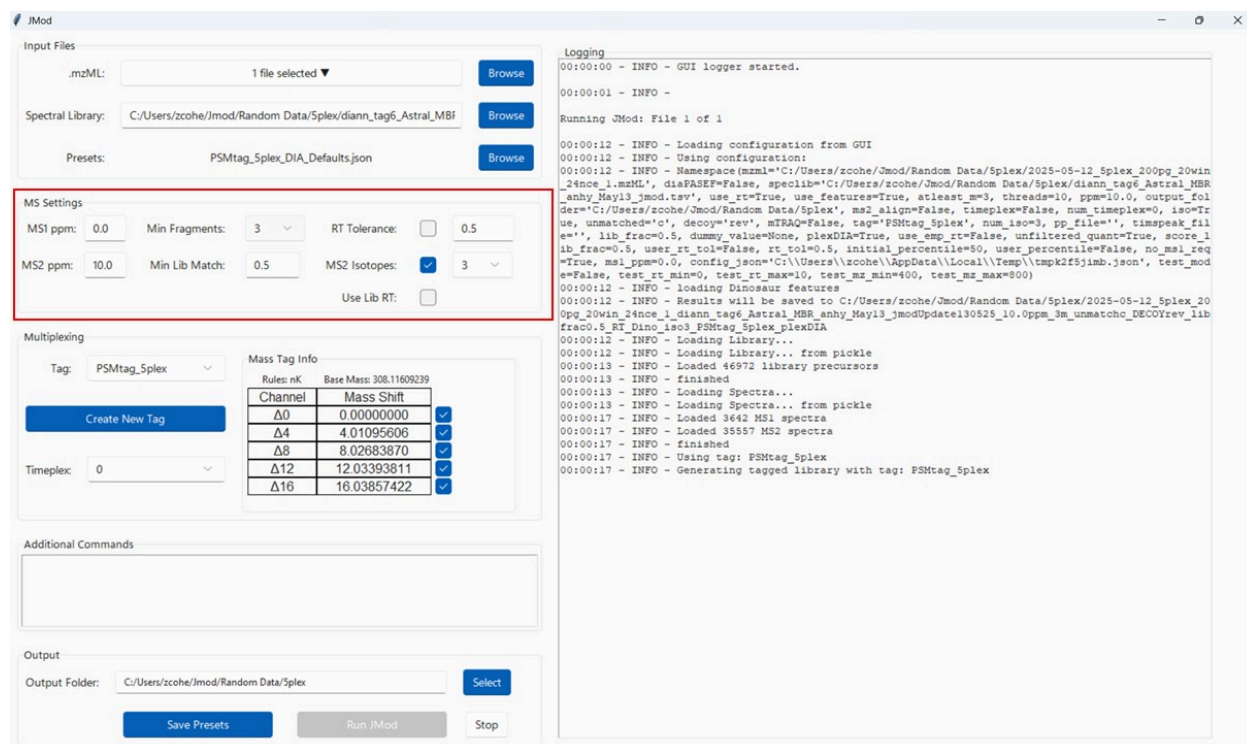
Channel	Mass Shift
$\Delta 0$	0.00000000
$\Delta 4$	4.01095606
$\Delta 8$	8.02683870
$\Delta 12$	12.03393811
$\Delta 16$	16.03857422

The 'Additional Commands' section is empty. The 'Output' section shows an 'Output Folder' set to 'C:/Users/zcohe/mod/Random Data/Splex' with a 'Select' button. At the bottom, there are 'Save Presets', 'Run JMod', and 'Stop' buttons. On the right, a 'Logging' window shows the following log entries:

```
00:00:00 - INFO - GUI logger started.
00:00:01 - INFO - 
Running JMod: File 1 of 1
00:00:12 - INFO - Loading configuration from GUI
00:00:12 - INFO - Using configuration:
00:00:12 - INFO - Namespace(mzml='C:/Users/zcohe/Jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1.mzML', dieFASTP=False, specLib='C:/Users/zcohe/Jmod/Random Data/Splex/diann_tag6_Astral_MBI_anhy_May13_jmod.tsv', use_rt=True, use_features=True, atleast_m=3, threads=10, ppm=10.0, output_folder='C:/Users/zcohe/Jmod/Random Data/Splex', ms2_align=False, timeplex=False, num_timeplex=0, iso=True, unmatched='c', decoy='rev', mTRAQ=False, tag='PSMtag_Splex', num_iso=3, pp_file='', timepeak_file='', lib_frac=0.5, dummy_value=None, plexDIA=True, use_emp_rt=False, unfiltered_quant=True, score_lib_frac=0.5, user_rt_tol=False, rt_tol=0.5, initial_percentile=50, user_percentile=False, no_msi_req=True, msi_ppm=0.0, config_json='C:/Users/zcohe/AppData/Local/Temp/tmpk2f5jmb.json', test_mode=False, test_rt_min=0, test_rt_max=10, test_ms_min=400, test_ms_max=800)
00:00:12 - INFO - Results will be saved to C:/Users/zcohe/Jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1_diann_tag6_Astral_MBI_anhy_May13_jmodUpdate130525_10.0ppm_3m_unmatchc_DECORrev_lib_frac0.5_RT_Dino_iso3_PSMtag_Splex_plexDIA
00:00:12 - INFO - Loading Library...
00:00:12 - INFO - Loading Library... from pickle
00:00:13 - INFO - Loaded 46972 library precursors
00:00:13 - INFO - finished
00:00:13 - INFO - Loading Spectra...
00:00:13 - INFO - Loading Spectra... from pickle
00:00:17 - INFO - Loaded 3642 MS1 spectra
00:00:17 - INFO - Loaded 35557 MS2 spectra
00:00:17 - INFO - finished
00:00:17 - INFO - Using tag: PSMtag_Splex
00:00:17 - INFO - Generating tagged library with tag: PSMtag_Splex
```

JMod takes one or multiple mzML files as input. JMod requires a spectral library for use. For multiplexed data with nonisobaric mass tags (such as PSMtag), the spectral library should contain peptides labelled with the lowest mass isotopologue (or “d0”). We recommend using empirical spectral libraries for optimal performance. For more on generating PSMtag libraries, check out [this blogpost](#). Optionally, a configuration file can be chosen from among a set of presets, or from past JMod runs. Configuration files can be found in the output folder of all JMod searches.

Step 2: Selecting MS related parameters



Various parameters of the JMod search can be modulated from the GUI. These include:

MS1 ppm	:	MS1 matching tolerance in parts-per-million. If 0, optimized value will be calculated
MS2 ppm	:	MS2 matching tolerance in parts-per-million
Min Fragments	:	Minimum number of fragments matched for precursor identification.
Min Lib Match	:	Minimum fraction of library intensity matched for precursor to be considered
RT Tolerance	:	If turned on, set an explicit retention time tolerance to be used.
MS2 Isotopes	:	If turned on, the number of isotopes used to search MS2 spectra
Use lib RT	:	Use library retention time instead of fine-tuning retention time

Step 3: Selecting Parameters for Multiplexing

Input Files

.mzML: 1 file selected

Spectral Library: C:/Users/zcohe/jmod/Random Data/Splex/diann_tag6_Astral_MBI

Presets: PSMtag_Splex_DIA_Defaults.json

MS Settings

MS1 ppm: 0.0 Min Fragments: 3 RT Tolerance: ☐ 0.5

MS2 ppm: 10.0 Min Lib Match: 0.5 MS2 Isotopes: ☒ 3

Use Lib RT: ☐

Multiplexing

Tag: PSMtag_Splex

Timeplex: 0

Mass Tag Info

Rules: nK Base Mass: 308.11609239

Channel	Mass Shift	
Δ0	0.00000000	☒
Δ4	4.01095606	☒
Δ8	8.02683870	☒
Δ12	12.03393811	☒
Δ16	16.03857422	☒

Additional Commands

Output

Output Folder: C:/Users/zcohe/jmod/Random Data/Splex

Logging

```
00:00:00 - INFO - GUI logger started.
00:00:01 - INFO -
Running JMod: File 1 of 1
00:00:12 - INFO - Loading configuration from GUI
00:00:12 - INFO - Using configuration:
00:00:12 - INFO - Namespace(mzml='C:/Users/zcohe/jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1.mzML', diaFASEP=False, specLib='C:/Users/zcohe/jmod/Random Data/Splex/diann_tag6_Astral_MBI_anhy_May13_3mod.tsv', use_rt=True, use_features=True, atleast_m=3, threads=10, ppm=10.0, output_folder='C:/Users/zcohe/jmod/Random Data/Splex', ms2_align=False, timeplex=False, num_timeplex=0, isoTr=us, unmatched='o', decoy='rev', mTRAQ=False, tag='PSMtag_Splex', num_iso3, pp_file='', timespeak_file='', lib_frac=0.5, dummy_value=None, plexDIA=True, use_esp_rt=False, unfiltered_quant=True, score_1ib_frac=0.5, user_rt_tol=False, rt_tol=0.5, initial_percentile=50, user_percentile=False, no_ms1_req=True, ms1_ppm=0.0, config_json='C:/Users/zcohe/AppData/Local/Temp/tmpk2f5jmb.json', test_mode=False, test_rt_min=0, test_rt_max=10, test_mz_min=400, test_mz_max=800)
00:00:12 - INFO - Loading Dinosaur features
00:00:12 - INFO - Results will be saved to C:/Users/zcohe/jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1_diann_tag6_Astral_MBI_anhy_May13_3modUpdate130525_10.0ppm_3m_unmatchc_DECORrev_lib_frac0.5_RT_Dino_iso3_PSMtag_Splex_plexDIA
00:00:12 - INFO - Loading Library...
00:00:12 - INFO - Loading Library... from pickle
00:00:13 - INFO - Loaded 46972 library precursors
00:00:13 - INFO - finished
00:00:13 - INFO - Loading Spectra...
00:00:13 - INFO - Loading Spectra... from pickle
00:00:17 - INFO - Loaded 3642 MS1 spectra
00:00:17 - INFO - Loaded 3557 MS2 spectra
00:00:17 - INFO - finished
00:00:17 - INFO - Using tag: PSMtag_Splex
00:00:17 - INFO - Generating tagged library with tag: PSMtag_Splex
```

Next, if multiplexing, either with nonisobaric mass tags or with timeplex, parameters for these can be selected. From a dropdown menu, select from premade nonisobaric tags, including diethyl, mTRAQ, and PSMtag. For label free data, set the tag to “None”. The channels and mass shifts for the tag will be displayed to the right. By selecting or unselecting the boxes next to each channel, data can be searched with this subset of channels only. Additionally, other tags can be created for the search with the “Create New Tag” button. Once created, these will populate in the tag dropdown menu for this run, as well as subsequent runs. If using timeplex, the number of time channels in use can be selected from the timeplex dropdown menu.

Step 4: Additional Commands

The screenshot shows the JMod GUI with various configuration options. The 'Additional Commands' section is highlighted with a red box. The 'Logging' panel on the right shows the execution log.

Input Files

.mzML: 1 file selected

Spectral Library: C:/Users/zcohe/jmod/Random Data/SpIex/diann_tag6_Astral_MBF

Presets: PSMtag_Splex_DIA_Defaults.json

MS Settings

MS1 ppm: 0.0 Min Fragments: 3 RT Tolerance: ☐ 0.5

MS2 ppm: 10.0 Min Lib Match: 0.5 MS2 Isotopes: ☒ 3

Use Lib RT: ☐

Multiplexing

Tag: PSMtag_Splex

Timeplex: 0

Mass Tag Info

Rules: nK Base Mass: 308.1169239

Channel	Mass Shift	
$\Delta 0$	0.00000000	☒
$\Delta 4$	4.01095606	☒
$\Delta 8$	8.02683870	☒
$\Delta 12$	12.03393811	☒
$\Delta 16$	16.03857422	☒

Additional Commands

Output

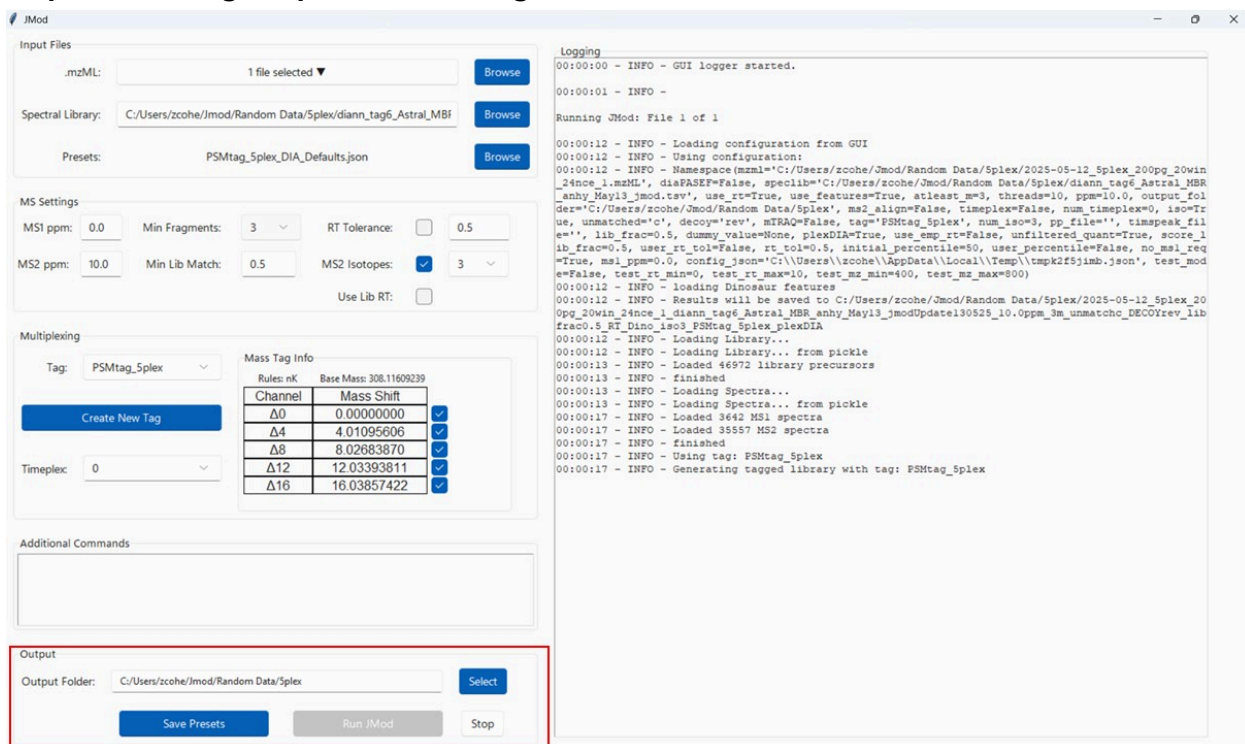
Output Folder: C:/Users/zcohe/jmod/Random Data/SpIex

Logging

```
00:00:00 - INFO - GUI logger started.
00:00:01 - INFO -
Running JMod: File 1 of 1
00:00:12 - INFO - Loading configuration from GUI
00:00:12 - INFO - Using configuration:
00:00:12 - INFO - Namespace (mmml="C:/Users/zcohe/jmod/Random Data/SpIex/2025-05-12_Splex_200pg_20win
24nce_1_mzML", diaPARSE=False, specLib="C:/Users/zcohe/jmod/Random Data/SpIex/diann_tag6_Astral_MBF
_anhy_May13_jmod.tsv", use_rt=True, use_features=True, atleast_mw=3, threads=10, ppm=10.0, output_fol
der="C:/Users/zcohe/jmod/Random Data/SpIex", ms2_align=False, timeplex=False, num_timeplex=0, iso=Tr
ue, unmatched="c", decoy="rev", mTRAQ=False, tag="PSMtag_Splex", num_iso=3, pp_file="", timespeak_fil
e="", lib_frac=0.5, dummy_value=None, plexDIA=True, use_esp_rt=False, unfiltered_quant=True, score_1
ib_frac=0.5, user_rt_tol=False, rt_tol=0.5, initial_percentile=50, user_percentile=False, no_ms1_req
=True, ms1_ppm=0.0, config_json="C:/Users/zcohe/AppData/Local/Temp/tmpk2f5jmb.json", Test_mod
e=False, test_rt_min=0, test_rt_max=10, test_mz_min=400, test_mz_max=800)
00:00:12 - INFO - loading Dinosaur features
00:00:12 - INFO - Results will be saved to C:/Users/zcohe/jmod/Random Data/SpIex/2025-05-12_Splex_20
0pg_20win_24nce_1_diann_tag6_Astral_MBF_anhy_May13_jmodUpdate130525_10.0ppm_3m_unmatchc_DECOYrev_lib
frac0.5_RT_Dino_1mod_PSMtag_Splex_plexDIA
00:00:12 - INFO - Loading Library...
00:00:12 - INFO - Loading Library... from pickle
00:00:13 - INFO - Loaded 46972 library precursors
00:00:13 - INFO - finished
00:00:13 - INFO - Loading Spectra...
00:00:13 - INFO - Loading Spectra... from pickle
00:00:17 - INFO - Loaded 3642 MS1 spectra
00:00:17 - INFO - Loaded 35557 MS2 spectra
00:00:17 - INFO - finished
00:00:17 - INFO - Using tag: PSMtag_Splex
00:00:17 - INFO - Generating tagged library with tag: PSMtag_Splex
```

While the GUI does not have a widget for every possible command line argument, these arguments can be written exactly as they are in the command line in the entry here and will be parsed the same way.

Step 5: Selecting Output and Running



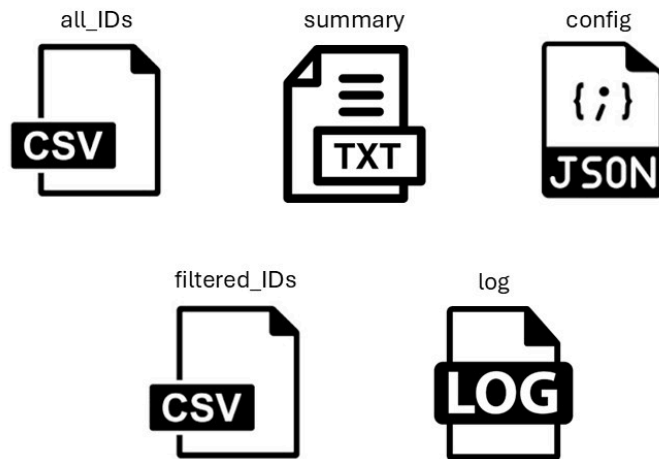
The screenshot displays the JMod software interface. The 'Output' section at the bottom is highlighted with a red box, showing the 'Output Folder' set to 'C:/Users/zcohe/jmod/Random Data/Splex'. The 'Logging' window on the right provides a detailed log of the search process, including file loading, configuration, and the generation of a tagged library.

Logging

```
00:00:00 - INFO - GUI logger started.
00:00:01 - INFO -
Running JMod: File 1 of 1
00:00:12 - INFO - Loading configuration from GUI
00:00:12 - INFO - Using configuration:
00:00:12 - INFO - Namespace(mzml='C:/Users/zcohe/jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1.mzML', diaFASEP=False, specLib='C:/Users/zcohe/jmod/Random Data/Splex/diann_tag6_Astral_MBR_anhy_May13_jmod.tsv', use_rt=True, use_features=True, atleast_m=3, threads=10, ppm=10.0, output_folder='C:/Users/zcohe/jmod/Random Data/Splex', ms2_align=False, timeplex=False, num_timeplex=0, iso=True, unmatched='o', decoy='rev', mTRAQ=False, tag='PSMtag_Splex', num_iso=3, pp_file='', timespeak_file='', lib_frac=0.5, dummy_value=None, plexDIA=True, use_esp_rt=False, unfiltered_quant=True, score_1ib_frac=0.5, user_rt_tol=False, rt_tol=0.5, initial_percentile=50, user_percentile=False, no_ms1_req=True, ms1_ppm=0.0, config_json='C:/Users/zcohe/AppData/Local/Temp/tmpk2f5jmb.json', test_mode=False, test_rt_min=0, test_rt_max=10, test_ms_min=400, test_ms_max=800)
00:00:12 - INFO - Loading Dinosaur features
00:00:12 - INFO - Results will be saved to C:/Users/zcohe/jmod/Random Data/Splex/2025-05-12_Splex_200pg_20win_24nce_1_diann_tag6_Astral_MBR_anhy_May13_jmodUpdate130525_10.0ppm_3m_unmatchc_DECOYrev_lib_frac0.5_RT_Dino_iso3_PSMtag_Splex_plexDIA
00:00:12 - INFO - Loading Library...
00:00:12 - INFO - Loading Library... from pickle
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00:00:17 - INFO - Loaded 3557 MS2 spectra
00:00:17 - INFO - finished
00:00:17 - INFO - Using tag: PSMtag_Splex
00:00:17 - INFO - Generating tagged library with tag: PSMtag_Splex
```

Select a folder for outputs to be placed in. Each .mzML file will have a unique folder within this containing its specific outputs. A configuration file for the current parameters can be saved beforehand using “Save Presets”. However, this configuration file will also be created automatically in the output folder. Selecting run will perform sequential JMod searches on each selected .mzML. Progress can be monitored via the log output on the right-hand side of the GUI, and outputs from each file are available as soon as it is finished running.

Step 6: Looking at Jmod Outputs



JMod outputs 5 important files (along with many others that can be used for quality control). Tabular results are contained in **all_IDs.csv** and **filtered_IDs.csv** (with **filtered_IDs** showing only peptides with $Q < 0.01$). Columns for these are described [hyperlink\[here\]](#). **Summary.txt** shows the number of proteins and precursors identified at 1% FDR, and for plexDIA additionally shows the number of precursors and proteins where the best channel is identified at 1% FDR. Additionally **Log.log** contains logged statements from the run, and **config.json** contains the parameters of the run, which can be used as a “preset” in the GUI, or to run a search from the command line.